

# Principia Orthogona

Volume Two: Contact Realization of Generative Transitions

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2026

*Dedicated to my children Vic (R.I.P.), Giulia, Alice (R.I.P.), Sarah (R.I.P.), and David.*

*Once tiny, always strong.*

## Contents

|          |                                                                     |          |
|----------|---------------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                                 | <b>3</b> |
| 1.1      | The problem: from geometric folds to dissipative dynamics . . . . . | 3        |
| 1.2      | Role of the dm3 toy model . . . . .                                 | 4        |
| 1.3      | Main results . . . . .                                              | 4        |
| 1.4      | Standing assumptions . . . . .                                      | 5        |
| <b>2</b> | <b>Contact Realization of the Fold Operator</b>                     | <b>5</b> |
| 2.1      | From Hamiltonian phase space to contact extension . . . . .         | 5        |
| 2.2      | The fold as a contact discontinuity . . . . .                       | 5        |
| 2.3      | Relation to the dm3 normal form . . . . .                           | 5        |
| 2.4      | Correspondence table . . . . .                                      | 6        |
| <b>3</b> | <b>Equivalence of <math>\kappa^*</math> and <math>\tau</math></b>   | <b>6</b> |
| 3.1      | Geometric threshold implies stochastic threshold . . . . .          | 6        |
| 3.2      | Stochastic threshold implies geometric threshold . . . . .          | 6        |
| 3.3      | The equivalence theorem . . . . .                                   | 7        |

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|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| <b>4</b> | <b>Explicit Verification in the dm3 Toy Model</b>   | <b>7</b>  |
| 4.1      | The system . . . . .                                | 7         |
| 4.2      | The verification theorem . . . . .                  | 7         |
| 4.3      | The system . . . . .                                | 8         |
| 4.4      | Threshold verification . . . . .                    | 8         |
| 4.5      | Normal form verification . . . . .                  | 9         |
| 4.6      | Stability radius . . . . .                          | 9         |
| <b>5</b> | <b>Singularity–Bifurcation Correspondence</b>       | <b>9</b>  |
| 5.1      | The four bifurcations of the toy model . . . . .    | 9         |
| 5.2      | Correspondence with Whitney singularities . . . . . | 9         |
| <b>6</b> | <b>Discussion</b>                                   | <b>10</b> |
| 6.1      | Summary of the three main results . . . . .         | 10        |
| 6.2      | Role in the trilogy . . . . .                       | 10        |
| 6.3      | Open problems . . . . .                             | 10        |

# Preface

This volume is the second in the Principia Orthogona series. Volume One [1] developed the singularity-theoretic and variational foundations of generative transitions: the operator sequence  $C \rightarrow K \rightarrow F \rightarrow U$ , the curvature threshold  $\kappa^*$ , the Whitney  $A_1$ – $A_3$  singularity classification, and a symplectic preservation theorem for the fold map.

The present volume constructs the explicit contact-geometric realization of those foundations. The central results are:

- a precise correspondence between the geometric fold operator  $F$  and the dm3 contact Hamiltonian dissipation (Section 2),
- a theorem establishing the equivalence of the curvature threshold  $\kappa^*$  and the embodiment threshold  $\tau$  (Section 3),
- explicit verification of the correspondence in the dm3 toy model (Section 4),
- a bifurcation analysis showing that the four dm3 bifurcations correspond to the singularity types of Volume One (Section 5).

Throughout, we take as established the results of Volume One [1], Generative Contact Mechanics [2], and the dm3 Toy Model [3]. This volume does not re-derive those results; it constructs the contact-geometric bridge between the geometric framework of Volume One and the dynamical framework of [2, 3].

Volume Three will instantiate the framework in specific domains: plasma reconnection, market volatility manifolds, and neural embedding geometry.

## 1 Introduction

### 1.1 The problem: from geometric folds to dissipative dynamics

Volume One established that generative transitions are localized geometric events classified by the Whitney  $A_1$ – $A_3$  hierarchy. The fold operator  $F$  was shown to act as a symplectic canonical transformation on  $T^*X$ , and the full transition  $\mathcal{G} = U \circ F \circ K \circ C$  was shown to be a piecewise-smooth symplectic map. What Volume One deliberately left open was the question of *post-fold stability*: once the fold has occurred and the unfolding  $U$  has selected a stable branch, what governs the long-term dissipative dynamics near that branch?

The Hamiltonian framework of Volume One cannot answer this question: Liouville’s theorem forbids attractors in symplectic systems on compact manifolds. The answer is contact geometry. The contact manifold  $M = X \times \mathbb{R}$  with contact form  $\alpha = dz - \lambda$  provides the correct geometric setting for dissipative dynamics with limit cycle attractors, stochastic stability, and variational structure. This is the framework of [2].

## 1.2 Role of the dm3 toy model

The dm3 Toy Model [3] is not an example among many, nor a numerical illustration of general theory. Its role is structural and methodological: it serves as a complete, explicit realization of the abstract contact-geometric framework of [2] and as the bridge that certifies the realizability of the geometric fold theory of Volume One.

Its purpose can be stated precisely:

*The dm3 Toy Model proves that the generative transition framework is not merely definable, but fully instantiable by an explicit, smooth, globally analyzable dynamical system.*

This has three consequences. First, the definitions do not depend on the toy model; no axiom is inferred from it and no theorem is tailored to fit it. The model is introduced after the theory is fully stated, to verify rather than to inspire. Second, the toy model converts abstract structure into computable structure: every operator, invariant, threshold, and bifurcation predicted by the theory can be checked by direct calculation. Third, the toy model certifies non-emptiness: without it, the axioms and operators could be mutually consistent but never jointly satisfiable. The existence of at least one concrete realization proves there are no hidden contradictions.

The toy model does not claim to be generic, to represent a physical system, or to capture all possible generative transitions. Its role is logical, not empirical.

## 1.3 Main results

**Theorem 1.1** (A: Contact realization of the fold). *The fold operator  $F$  of Volume One is the piecewise-smooth, pre-contact limit of the dm3 operator  $\mathcal{A}_{\text{dm3}} = \phi^{T^*/4}$  of [2]. Under the contact extension  $M = X \times \mathbb{R}$ , the impulsive momentum jump  $p^+ - p^- = \mu n$  at the fold corresponds to the contact Hamiltonian correction  $H_{\text{diss}} = -\gamma V e^{-\beta z}$  in the regularized limit  $\beta \rightarrow \infty$ .*

**Theorem 1.2** (B: Threshold equivalence). *Under the generative model of Volume One and the dm3 realization of [2]:*

$$|\kappa| \uparrow \kappa^* \iff \mu_{\text{max}} < 0 \iff \tau = \sqrt{c/\kappa_{\text{noise}}} \in (0, \infty).$$

*The curvature threshold  $\kappa^*$  and the embodiment threshold  $\tau$  are two parameterizations of the same event: the onset of transverse stability in the post-fold dissipative system.*

**Theorem 1.3** (C: Singularity–bifurcation correspondence). *The four bifurcations of the dm3 toy model [3] correspond to the Whitney singularity types of Volume One under the projection  $M = S \times \mathbb{R} \rightarrow S$ .*

## 1.4 Standing assumptions

**Assumption 1.1** (Volume One framework). *The operator sequence  $C \rightarrow K \rightarrow F \rightarrow U$ , the threshold  $\kappa^*$ , and all results of [1] hold.*

**Assumption 1.2** (dm3 framework). *The dm3 system, contact manifold  $M = S \times \mathbb{R}$ , and all results of [2, 3] hold.*

## 2 Contact Realization of the Fold Operator

### 2.1 From Hamiltonian phase space to contact extension

In Volume One, the fold is realized in  $T^*X$  as a symplectic discontinuity. At fold point  $s_0$ :

$$p(s_0^+) - p(s_0^-) = \mu n(s_0),$$

generated by  $S(\gamma) = \mu \Theta(\kappa(\gamma) - \kappa^*)$  ([1], §12). The fold map  $\mathcal{F} : (\gamma, p) \mapsto (\gamma, p + \mu n)$  preserves  $\omega = d\gamma \wedge dp$  ([1], Theorem 11.1).

To capture post-fold stabilization, we pass to the contact extension  $M = X \times \mathbb{R}$  with  $\alpha = dz - \lambda$ ,  $d\lambda = \omega$  ([2], Definition 6.1).

### 2.2 The fold as a contact discontinuity

**Proposition 2.1** (Regularization of the fold generator). *The contact dissipation  $H_{\text{diss}}(x, z) = -\gamma V(x)e^{-\beta z}$  is a smooth regularization of  $S(\gamma) = \mu \Theta(\kappa(\gamma) - \kappa^*)$ : as  $\beta \rightarrow \infty$  and  $z \rightarrow 0^+$ , the correction  $-\gamma \nabla V e^{-\beta z}$  concentrates near  $\Gamma = \{V = 0\}$  and mimics the threshold activation of  $\Theta$  at  $\kappa = \kappa^*$ .*

*Proof sketch.* For large  $\beta$ ,  $e^{-\beta z} \rightarrow 0$  away from  $z \approx 0$ , so  $H_{\text{diss}}$  is negligible in the post-embodiment regime. Near  $z = 0$ ,  $V \approx 0$ , the correction provides an impulsive push toward  $\Gamma$  of magnitude  $\mu = \gamma \|\nabla V\|$ . In the limit  $\beta \rightarrow \infty$ , this recovers step-function activation, corresponding to  $S$ .  $\square$

The contact dissipation has three structural properties: (i)  $H_{\text{diss}}|_{\Gamma} = 0$  (invariant set preserved); (ii) off  $\Gamma$ :  $H_{\text{diss}} < 0$  (contraction); (iii)  $e^{-\beta z}$  weakens dissipation as action accumulates (embodiment threshold).

### 2.3 Relation to the dm3 normal form

By [2] (Theorem C), every dm3 system near  $\Gamma$  is locally contact-diffeomorphic to

$$\dot{\rho} = \mu_{\max}(1 - e^{-\beta z})\rho + O(\rho^2), \quad (2.1)$$

$$\dot{\theta} = \omega + O(\rho), \quad (2.2)$$

$$\dot{z} = \omega - |\mu_{\max}| \rho^2 e^{-\beta z} + O(\rho^3). \quad (2.3)$$

In the  $C \rightarrow K \rightarrow F \rightarrow U$  sequence:  $\rho \rightarrow 0$  is the unfolding  $U$ ;  $\mu_{\max} < 0$  is the imprint of having crossed  $\kappa^*$ ;  $z$  growing records accumulated dissipation from the fold.

## 2.4 Correspondence table

| <i>Volume One (geometric)</i>    | <i>GCM / dm3 (contact)</i>                      |
|----------------------------------|-------------------------------------------------|
| Curvature threshold $\kappa^*$   | Onset of transverse contraction                 |
| Fold impulse $p^+ - p^- = \mu n$ | Contact correction $H_{\text{diss}}$            |
| Rank-1 Jacobian loss             | Dissipative contact normal form                 |
| Unfolding $U$                    | Gradient flow to $\Gamma$ via $U_3$ [2]         |
| Distributional generator $S$     | Regularized $H_{\text{diss}}$ (Proposition 2.1) |

Table 1: Correspondence between Volume One and the contact realization. This proves Theorem A.

## 3 Equivalence of $\kappa^*$ and $\tau$

### 3.1 Geometric threshold implies stochastic threshold

**Lemma 3.1** (Fold activation produces hyperbolicity). *If  $K$  drives curvature to  $\kappa^*$ ,  $F$  is rank-1, and  $U$  selects a nondegenerate branch, then  $\Gamma$  is hyperbolic with  $\mu_{\max} < 0$  and  $\dot{V} \leq -cV$  for some  $c > 0$ .*

*Proof sketch.* Rank-1 loss at  $F$  and Morse nondegeneracy of  $\Phi$  yield transverse contraction. Floquet theory gives  $\mu_{\max} < 0$  ([1], Theorem 3.3; [2], Proposition 3.2).  $\square$

**Theorem 3.2** (Geometric implies stochastic threshold). *Under Lemma 3.1,  $\mathcal{L}V \leq -cV + \kappa_{\text{noise}} \|\sigma\|^2$  and  $\tau = \sqrt{c/\kappa_{\text{noise}}} \in (0, \infty)$ .*

*Proof.*  $\dot{V} \leq -cV$  plus the Itô correction  $\frac{1}{2} \|\sigma\|^2 \|\text{Hess } V\|$  gives  $\mathcal{L}V \leq -cV + \kappa_{\text{noise}} \|\sigma\|^2$  with  $\kappa_{\text{noise}} = \frac{1}{2} \sup \|\text{Hess } V\|$ . Then  $\tau = \sqrt{c/\kappa_{\text{noise}}} > 0$  ([2], Definition 3.4).  $\square$

### 3.2 Stochastic threshold implies geometric threshold

**Lemma 3.3** (Finite  $\tau$  implies transverse contraction). *If  $\mathcal{L}V \leq -cV + \kappa_{\text{noise}} \|\sigma\|^2$  with  $c > 0$ , then  $\mu_{\max} < 0$  and  $\dot{V} \leq -cV$  near  $\Gamma$ .*

*Proof.* Exponential decay of  $\mathbb{E}[V(X_t)]$  forces  $\mu_{\max} < 0$  by stochastic stability theory ([2], Propositions 3.2 and 3.4).  $\square$

**Theorem 3.4** (Stochastic implies geometric threshold). *If  $\tau \in (0, \infty)$ , then the trajectory must have crossed  $\kappa^*$  and undergone a rank-1 fold.*

*Proof.* By Lemma 3.3,  $\mu_{\max} < 0$ , so a hyperbolic attracting cycle exists. Suppose  $|\kappa| < \kappa^*$  everywhere. Then [1] (Invariant 7.5) gives injectivity before threshold: no rank loss, no mechanism for a hyperbolic attracting cycle. Contradiction. So  $\kappa = \kappa^*$  must be reached.  $\square$

### 3.3 The equivalence theorem

**Theorem 3.5** (Threshold equivalence).

$$|\kappa| \uparrow \kappa^* \iff \mu_{\max} < 0 \iff \tau \in (0, \infty).$$

$\kappa^*$  is the geometric precursor of  $\tau$ : curvature accumulation creates the conditions under which stochastic stability becomes meaningful. This proves Theorem B.

*Proof.* Forward: Theorem 3.2. Backward: Theorem 3.4. Middle:  $\tau > 0 \iff c > 0 \iff \mu_{\max} < 0$  (Lemma 3.3).  $\square$

*Remark 3.1* (Scope). The equivalence is local to the fold neighborhood and the post-fold tubular neighborhood of  $\Gamma$ . The exclusion of  $A_k$  ( $k \geq 4$ ) in Volume One (Morse condition) aligns with contact normal form rigidity in [2] (Theorem C).

## 4 Explicit Verification in the dm3 Toy Model

### 4.1 The system

### 4.2 The verification theorem

Before exhibiting the explicit computations, we state the logical content of the entire verification as a single theorem.

**Theorem 4.1** (Verification of GCM by the dm3 Toy Model). *There exists an explicit smooth dynamical system on a contact manifold—the dm3 toy model of [3]—in which all of the following are verified by direct computation:*

- (1) **Axiom consistency.** *All eight dm3 axioms of [2] are satisfied simultaneously. The axiom class is non-empty.*
- (2) **Contact structure.** *The system lives on an explicit contact manifold with  $\alpha = dz - r^2 d\theta$ ; all contact identities hold by direct verification.*
- (3) **Contact normal form.** *The system is explicitly contact-diffeomorphic to the universal normal form of Theorem C of [2] with  $(\mu_{\max}, \omega, \beta) = (-2, 1, 1)$ . The normal form is non-empty.*
- (4) **Operator algebra closure.** *Every operator ( $g$ -,  $L$ -,  $R$ -,  $U$ -,  $B$ -operators) is instantiated explicitly and preserves the dm3 axioms. The full grammar closes.*

- (5) **Stochastic stability.**  $\tau = 2$  is computed in closed form. The stationary Fokker–Planck measure is Gaussian, concentrating on  $\Gamma$  for  $\sigma < \tau$  and spreading for  $\sigma > \tau$ .
- (6) **Global dynamics.** The global attractor is  $\Gamma_{12}$ ; all invariant sets are identified; all four predicted bifurcations are exhibited.

*Proof.* Items (1)–(6) are verified in [3] by direct computation: (1) §2.4; (2) Prop. 2.2; (3) §7; (4) §3; (5) §5 and Thm. D; (6) §§4,6 and Thm. A.  $\square$

*Remark 4.1* (Logical status of the verification). The dm3 Toy Model  $\mathcal{D}_0$  provides a constructive verification of GCM in the strongest possible sense: not by example or illustration, but by explicit realization. All abstract components of GCM—axioms, geometry, operators, thresholds, normal forms, stochastic predictions, and global dynamics—are shown to coexist within a single smooth contact-geometric system.

In one sentence: *the dm3 Toy Model is a complete verification witness for Generative Contact Mechanics, demonstrating that the entire abstract framework is jointly realizable, computable, and dynamically consistent.*

The toy model does not claim to be generic or to represent a physical system. Its role is logical, not empirical.

### 4.3 The system

On  $M = \mathbb{R}_{>0}^2 \times \mathbb{R}$  with polar coordinates  $(r, \theta, z)$ :

$$\dot{r} = r(1 - r^2) + 2(r - 1)e^{-z}, \quad (4.1)$$

$$\dot{\theta} = 1, \quad (4.2)$$

$$\dot{z} = r^2 - 2(r - 1)^2 e^{-z}. \quad (4.3)$$

Limit cycle:  $\Gamma = \{r = 1\}$ ,  $T^* = 2\pi$ . Contact form:  $\alpha = dz - r^2 d\theta$  ([3], Proposition 2.2).

### 4.4 Threshold verification

**Proposition 4.2** (Explicit threshold values).  $\mu_{\max} = -2$ ,  $\kappa_{\text{noise}} = 1$ ,  $\tau = 2$ . The transverse eigenvalue  $\lambda(z) = -2(1 - e^{-z})$  satisfies:  $\lambda(0) = 0$  (neutral, pre-embodiment),  $\lambda(z) < 0$  for  $z > 0$  (attracting, post-embodiment),  $\lambda(z) \rightarrow -2$  as  $z \rightarrow \infty$  (full dm3 rate).

*Proof.* Linearize at  $r = 1$ :  $\dot{\rho} = -2(1 - e^{-z})\rho + O(\rho^2)$ . Generator:  $\mathcal{L}V = -4V(1 - e^{-z}) + \sigma^2$ , giving  $c \rightarrow 4$ ,  $\kappa_{\text{noise}} = 1$ ,  $\tau = 2$  ([3], Lemma 2.1 and Proposition 2.4).  $\square$

The neutral stability at  $z = 0$  is the mathematical content of the embodiment threshold: the orbit earns its stability by accumulating action. The crossing  $z = 0 \rightarrow z > 0$  in the contact variable corresponds exactly to the curvature crossing  $\kappa \rightarrow \kappa^*$  in Volume One.



## 4.5 Normal form verification

**Proposition 4.3** (Contact normal form). *In coordinates  $(\rho, \theta, z)$  with  $\rho = r - 1$ , the system takes the form*

$$\dot{\rho} = -2(1 - e^{-z})\rho + O(\rho^2), \quad \dot{\theta} = 1 + O(\rho), \quad \dot{z} = 1 - 2\rho^2 e^{-z} + O(\rho^3),$$

*the contact normal form of [2] (Theorem C) with  $(\mu_{\max}, \omega, \beta) = (-2, 1, 1)$ .*

## 4.6 Stability radius

$$\varepsilon_0 = \frac{|\mu_{\max}|}{2(1 + \sup_{\Gamma} \|\text{Hess } V\|)} = \frac{2}{2(1 + 2)} = \frac{1}{3}.$$

# 5 Singularity–Bifurcation Correspondence

## 5.1 The four bifurcations of the toy model

From [3] (Theorem C):

- (i) **Contact Hopf** at  $\gamma = e^{z_0}$ : limit cycle loses stability, new cycle bifurcates.
- (ii) **Saddle-node** at  $\eta \approx 0.15$ : two cycles collide, multi-orbit structure collapses.
- (iii) **Neimark–Sacker** at detuning  $|\Delta| = \Delta^*$ : resonant orbit loses stability, 2-torus bifurcates.
- (iv) **Slow-fast crossover** at  $\beta = \beta^*$ : smooth transition between contact and classical regimes.

## 5.2 Correspondence with Whitney singularities

**Proposition 5.1** (Singularity–bifurcation correspondence). *Under projection  $M = S \times \mathbb{R} \rightarrow S$ :*

| <i>Singularity</i>  | <i>Codimension</i> | <i>dm3 bifurcation</i>       |
|---------------------|--------------------|------------------------------|
| $A_1$ (fold)        | 0                  | Contact Hopf and saddle-node |
| $A_2$ (cusp)        | 1                  | Neimark–Sacker               |
| $A_3$ (swallowtail) | 2                  | Slow-fast crossover          |

*Higher singularities excluded: in Volume One by the Morse condition; in [2] by contact normal form rigidity (Theorem C).*

*Proof sketch.*  $A_1$ : rank-1 loss in one transverse direction — mechanism of both Hopf (radial stability loss) and saddle-node (radial collision).  $A_2$ : rank-1 loss with second-order tangency — matches Neimark–Sacker (second angular direction destabilizes at resonance).  $A_3$ : rank-1 with third-order tangency — matches slow-fast crossover (third-order change in  $z$ -direction).  $\square$

This proves Theorem C.

## 6 Discussion

### 6.1 Summary of the three main results

- (1) The fold operator  $F$  is the geometric precursor of the dm3 contact Hamiltonian  $H_{\text{diss}}$ ; the distributional generator  $S$  is regularized by  $H_{\text{diss}}$  in the limiting sense of Proposition 2.1 (Theorem A).
- (2)  $\kappa^*$  and  $\tau$  are equivalent: each is finite and positive if and only if the other is, both detecting the onset of transverse stability (Theorem B).
- (3) The four dm3 bifurcations correspond to Whitney  $A_1$ – $A_3$  types, completing the singularity-theoretic classification of the toy model dynamics (Theorem C).

### 6.2 Role in the trilogy

- **Volume One:** why folds must occur; local geometric classification.
- **Volume Two:** what stable structures exist after folds; contact-geometric realization.
- **Volume Three:** domain-specific instantiations (plasma, markets, neural geometry).

### 6.3 Open problems

- (1) **Global equivalence.** Theorem B is local. A global version requires showing every  $\tau$ -stable dm3 system arises from a fold transition globally on  $X$ .
- (2) **Higher resonances.** Systematic treatment of  $k:m$  correspondence between higher  $A_k$  singularities and higher resonances.
- (3) **Volume Three.** Domain-specific  $(X, g, \Phi)$  with explicit computation of  $\kappa^*$  and  $\tau$  from data.

## Acknowledgments

The author acknowledges with gratitude the foundational influence of the teachings of Paramahansa Nithyananda and the yogic scriptural tradition from which this work derives. The mathematical formalization presented here is understood by the author as a translation of principles encountered through that tradition.

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